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MyDesign Portfolio

Element F Initial Template - AT - 3 points

How to use this template

Students: The template below is one way to assess whether your proposed solution is viable and realistic, but the specific assignment given by your instructor may require a different structure or dictate different details. If you make a copy of this template to edit, be sure to delete the instructions.

Element F: Consideration of Design Viability

Judgment about Design Viability (F1)

"Design viability" is the capacity of the proposed solution to solve the problem. Design viability refers to how your design decisions are tethered to your design requirements. For example, if one of your design requirements is that the total cost of the prototype cannot exceed X, then you could present a table of costs for each component of your proposed design solution as justification for why your design is likely to meet that requirement.

"Credible evidence" means it shouldn't just be your hunch that this will work. Be creative about how you validate the viability of your design. You should go beyond just having a stakeholder or expert reviewer say that the design seems viable. For example, you could point to evidence that a sub-system already functions, or you demo a proof of concept, or you've discussed with an expert that said, "I've seen something similar work in this other related situation...", etc.

Use as many rows as needed. Remember, this chart below may help you organize your ideas, but you will need to actually execute the "credible evidence."

Design Viability Description	Connection to Design Requirement from Element C	Credible Evidence?

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<i>Example: The solution is inexpensive</i>	<i>Example: Design does not cost more than \$X</i>	<i>Example: Create a table of costs for each component of the solution.</i>
The solution can be made with little to no cost to the school	School has no money to pay for this so we must use materials we already have	TPU - 20\$ per 100 millimeters Velcro strap - 1\$ per
The solution does not damage the floors and doesn't leave any marks	No scratching, staining or any type of marks will be made to the building	Test the solution and take before and after photos to show no damage to the floor
The solution can be put on in under 1-2 minutes	Must be quick for athletes to use	Time multiple athletes putting on the product over their cleats " Noah said anything under 2 minutes is acceptable"
The solution improves traction on the tile in the fine arts	Must stop slipping and add traction while walking	Do a traction test. see how far the cleats slide by themselves compared to a cleat with the cover on it record the results
The solution can be put on in one motion without removing the cleat	Must be easy to put on without the cleat taken off	The product stretches or clips on to easily fit on the shoe. Ask athletes to try it and make sure it is one motion
The solution is durable and can last a full year	Must not need to be replaced mid season	Do stress tests, lots of walking on tile, bending, being thrown ect. Make sure it stays intact. Compare the different



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		materials.
The solution is light weight and portable	Must be easy to carry.	

